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Virtual Gold (Hybrid Architecture) x CMU Capstone: Weekly Sync

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Attachments Virtual Gold (Hybrid Architecture) x CMU Capstone: Weekly Sync

Summary

Project setup and architecture planning with testing strategies and midterm preparation.

Capstone Project Agenda Setup
Role distribution and system architecture established. Evaluation dataset defined for testing.

System Architecture And Testing
Nine component architecture presented. Preliminary testing recommended using open source models.

Midterm Presentation Preparation
Faculty presentation structure discussed. Meeting concluded with administrative updates.

Decisions

Aligned

- **Preliminary testing of local model architecture** The team agreed to conduct preliminary testing of the local model and architecture using cloud-hosted or local models before next week.
- **Phased testing approach for redaction service** The participants aligned on a phased testing approach, starting with isolated functional testing of the redaction service before integration with VG08.
- **English-only input language scope for service** The group agreed to restrict input language and documents to English for the current phase.
- **Faculty presentation initial implementation structure** The team agreed to present their initial API evaluation as an initial implementation in the faculty presentation to demonstrate project progress.

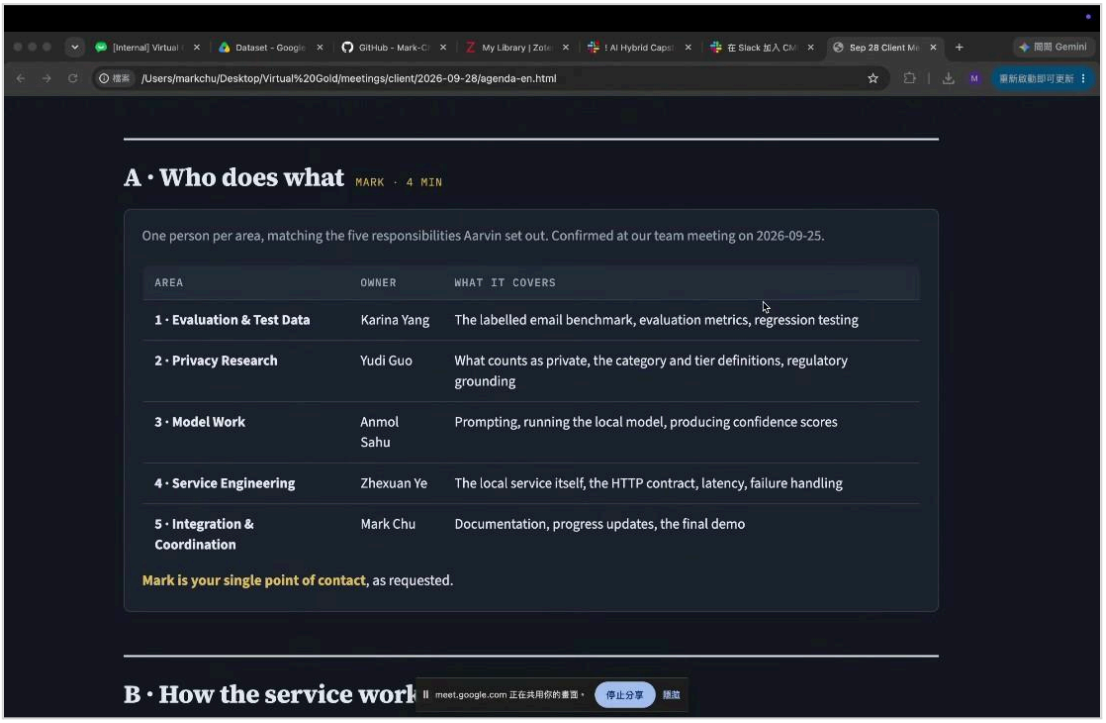
Next steps

- ☐ [Urte Jesina] Create Workspace: Create a new workspace to facilitate team communication.
- ☐ [The group] Test Architecture: Execute preliminary tests on the sensitivity classifier and redaction workflow using cloud services or local model instances.
- ☐ [Mark Chu] Share API Contract: Email the drafted API contract document to the team for review and feedback.
- ☐ [Alex, Aarvin George] Communicate Testing Plan: Align on the testing phases for the project and send a confirmation of the plan to the team.
- ☐ [Mark Chu] Share tables: Send the tables for information categories to the group for verification against project scenarios.
- ☐ [The group] Run local service: Implement and run the sensitive information detection service on a local system to test performance.
- ☐ [The group] Prepare presentation: Include the initial evaluation and test results as part of the midterm presentation next Monday.
- ☐ [Mark Chu] Upload scope: Provide the updated scope of work document.

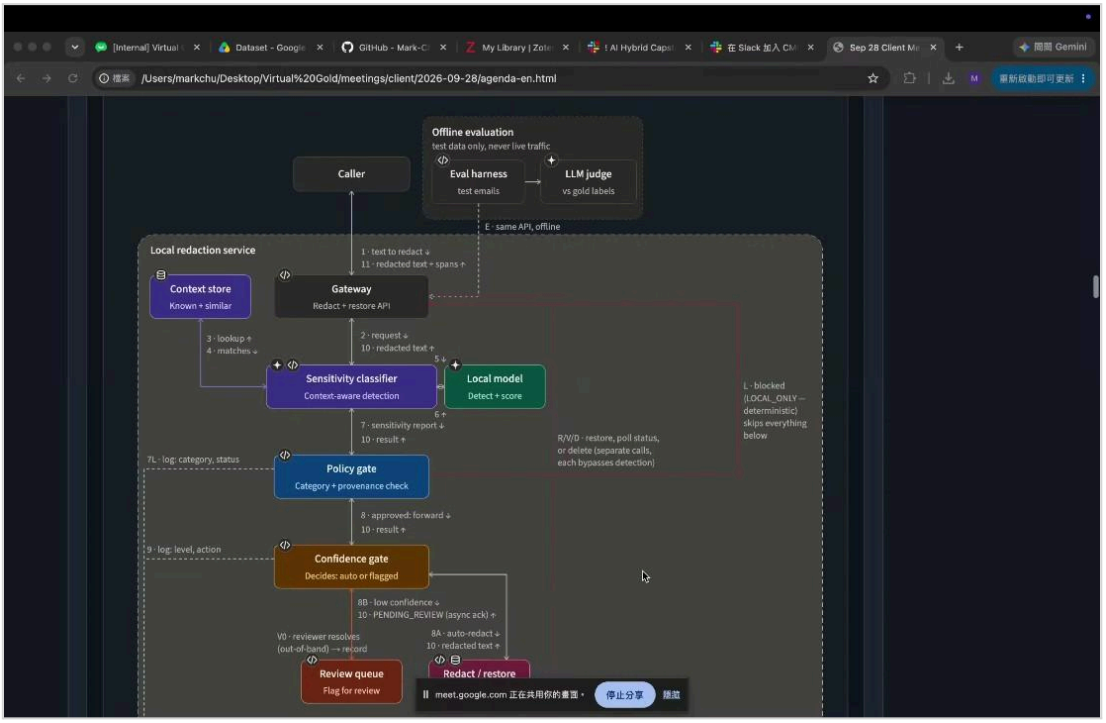
- [ ] [Mark Chu] Create workspace: Establish a new project workspace and share the link via Google Drive.

Details

- **Slack and Google Drive Setup**: Urte Jesina attempted to create and share a new Slack workspace link for the team, confirming that a paid plan was active despite previous error prompts. Meanwhile, Mark Chu inquired about Google Drive usage, noting that meeting notes and documents had been uploaded for Alex, Aarvin, Inderpal, and Arvin to review, which Urte Jesina confirmed receiving notifications for. Anmol Sahu successfully joined the new Slack workspace.
- **Capstone Project Agenda and Division of Responsibilities**: Mark Chu outlined the meeting agenda, covering five main areas: role distribution, system architecture, security module, evaluation dataset, and general discussion items such as GPU resources, NDAs, and upcoming midterm presentations. Responsibilities were assigned with Karina Yang handling evaluation and test data, Yudi Guo managing privacy research, Anmol Sahu and Zhexuan handling model work and architecture, and Mark Chu coordinating integration. Aarvin George clarified that these designations were meant to provide direction rather than strict boundaries, advising the team to avoid rabbit holes and aim for well-rounded contributions.

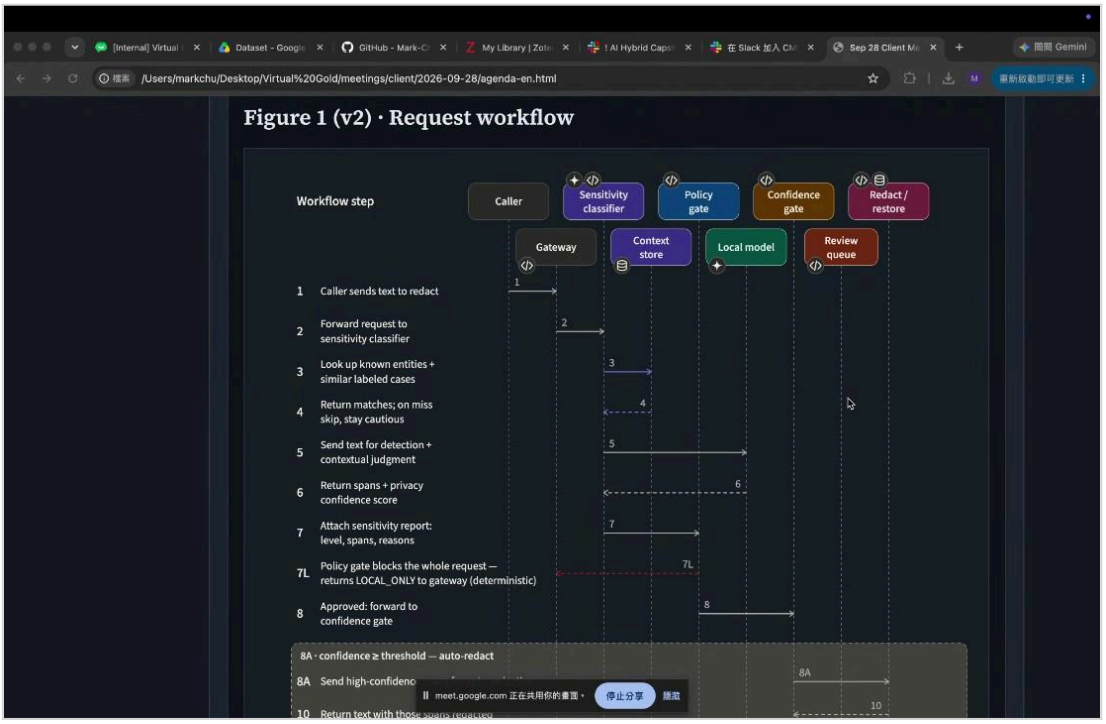


- **Evaluation Dataset and Service Engineering Clarification**: Inderpal Bhandari requested a walkthrough of the five work areas, prompting Karina Yang to explain that the evaluation dataset consists of 100 manually labeled emails with sensitive items marked to serve as an answer key for tracking whether the redaction service improves or degrades over time. Alex questioned the scope of service engineering, and Aarvin George clarified that it involves architecture design and building harnesses around the local model to serve as a screening layer between incoming user requests and core orchestration.
- **Proposed System Architecture Overview**: Mark Chu presented the proposed nine-component architecture, explaining that incoming text flows from the gateway to a sensitivity classifier, which checks a global cross-session context store of known entities and uses a local model to detect and score potentially sensitive spans. The sensitivity report passes to a policy gate containing rules that categorize requests (such as confidential, health, financial, or credential) to make a deterministic routing decision. Requests marked local-only return directly to the gateway, while others proceed to a confidence gate where high-confidence spans are automatically redacted and low-confidence spans are sent to a human review queue, with all gates generating audit logs.

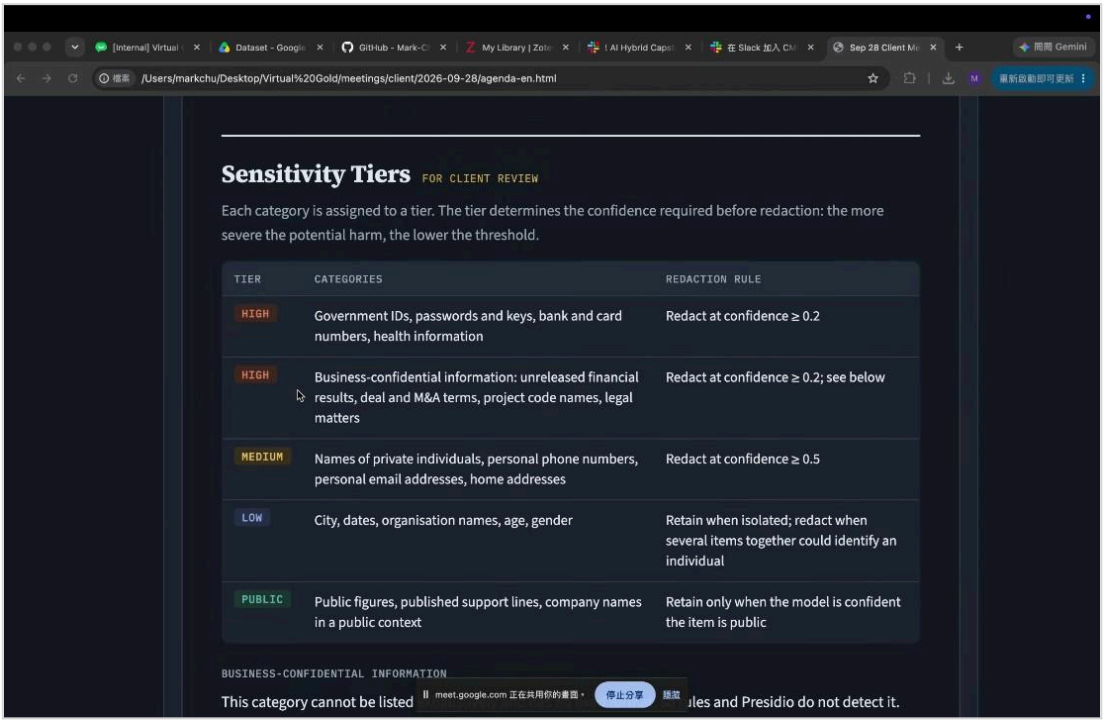


- **Testing Feasibility and API Contract**: Addressing questions from Alex regarding whether the architecture has been tested, Mark Chu stated that testing has not yet occurred due to ongoing development of benchmarks and architecture. Mark Chu noted that a drafted API contract uses structured inputs rather than plain text, exposing endpoints for sensitivity detection, placeholder restoration, and health probes. Alex and Aarvin George advised the team to perform preliminary testing as soon as possible over the web using open-source models to gauge architecture viability.
- **Pre-Sandbox Testing Recommendations**: Anmol Sahu raised a concern about needing a sandbox environment while restricted to personal laptops, but Inderpal Bhandari, Alex, and Aarvin George clarified that the team could perform preliminary testing over the web using cloud-hosted open-source models or local tools like Ollama with models such as Gemma before setting up a formal sandbox. Anmol Sahu acknowledged this approach, noting they could utilize free API keys from services like Groq for initial validation.
- **Detailed Workflow and Placeholder Pseudonymization**: Anmol Sahu detailed the workflow steps, explaining that the sensitivity classifier checks the context store, sends unmatched text to the local model with a system prompt, and receives sensitive spans and confidence scores. Anmol Sahu proposed using a cryptographic hash function like MD5 or pseudonymization to encrypt specific sensitive spans into placeholders before sending text out, and decrypting responses upon return using a service-held key. Following

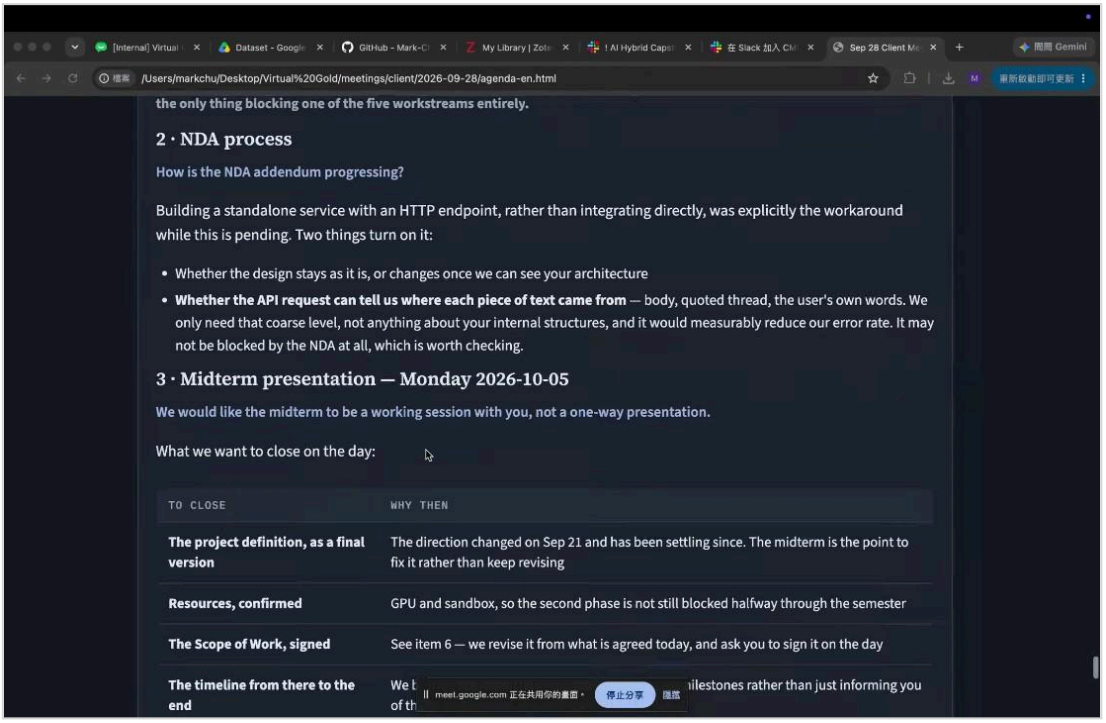
confusion from Alex and Aarvin George regarding the necessity and placement of encryption, the team discussed refining whether encryption or simpler pseudonymization is required for redacted text.



- **End-to-End Testing and Phased Implementation Plan**: Alex and Aarvin George emphasized the need for end-to-end testing to ensure the full chat-like system properly routes information and prevents sensitive data leaks to cloud APIs. Aarvin George and Alex agreed on a phased testing approach where Phase 1 involves isolating the redaction service to verify text and document processing locally, and Phase 2 involves integrating the HTTP server with Aarvin George's VG08 service to test live workflows. Anmol Sahu confirmed understanding of this standalone-to-integrated validation path for processing inputs such as state tax forms.
- **Sensitivity Classifier Methodology and Language Scope**: Yudi Guo outlined the methodology for the sensitivity classifier, which utilizes regex patterns for personal information like Social Security Numbers or government IDs, open-source named entity recognition, and local model detection mapped across categories with tiered sensitivity thresholds. Anmol Sahu confirmed with Aarvin George and Inderpal Bhandari that the project will exclusively target English-language text and documents for the time being.



- **Technical Support and Midterm Presentation Preparation**: Mark Chu discussed utilizing Slack for ongoing technical alignment and troubleshooting with Alex and Aarvin George rather than waiting for weekly meetings. Additionally, the team discussed preparing for next week's midterm presentation scheduled for Monday from 6:00 p.m. to 7:00 p.m., where they will present the refined scope of work, basic requirements, delivery documents, and system architecture.



- **Project Scope and Deliverables Management**: The team discusses the potential problem of scope creep while working on Virtual Gold. Aarvin George advises keeping project deliverables limited and achievable because larger scopes can become daunting. Mark Chu agrees with this approach, and Anmol Sahu notes that their team will hold a separate discussion to finalize these manageable goals.
- **Faculty Presentation Recommendations**: Inderpal Bhandari addresses the structure of the upcoming faculty presentation, suggesting that the team showcase their initial evaluation—which utilizes free APIs to test ideas—as a solid early implementation. Inderpal Bhandari points out that this approach demonstrates architectural flow and tangible progress before presenting deliverables for upcoming weeks. Mark Chu expresses appreciation for the feedback, and Anmol Sahu confirms that everyone on the team will participate in the midterm presentation.
- **Midterm Presentation Schedule and Attendance**: The team discusses the scheduling of the midterm presentation, which is set for next Monday, the 5th, from 6 p.m. to 7 p.m. Inderpal Bhandari states they have a conflict due to a West Coast client meeting on that day, but confirms that Alex and Aarvin George will be able to represent. Alex verifies that the one-hour time slot works for their schedule.
- **Resource Request Process**: Mark Chu raises a question regarding how to request resources needed for development functions. Anmol Sahu suggests

postponing this discussion, noting that previous questions have already been addressed. Inderpal Bhandari clarifies the workflow for acquiring resources, stating that once a need is discussed and confirmed with Arvin and Alex, the team should notify the designated individual to secure Inderpal Bhandari's approval before proceeding.

- **Project Documentation and Meeting Conclusion**: Mark Chu provides updates on project administration, stating that they have uploaded the new scope of work, created a workspace, and provided a Google Drive link where future meeting notes and documents will be stored. Inderpal Bhandari and Mark Chu acknowledge the strong progress made, and participants conclude the meeting.

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